Reg. No. | | | | | | | | | | | | |

BCACACN 403

Fourth Semester B.C.A. Degree Examination, June/July 2025
(NEP – 2020) (2022 – 23 Batch Onwards)
OPERATING SYSTEM CONCEPTS

Time : 2 Hours

Max. Marks : 60

Note : *Answer* ***any six*** *questions from Part –* ***A*** *and* ***any one*** *full question from* ***each*** *Unit of Part –* ***B****.*

PART – A **(6×2=12)**

1. a) Define an Operating System.

   b) Give any four attributes of file.

   c) Define logical and physical address.

   d) What is memory-management unit ?

   e) What is PCB ? List its component.

   f) Define CPU and I/O Burst.

   g) Mention the two methods of handling the deadlock.

   h) What is semaphore ?

PART – B

Unit – I

2. a) Write a note on following :
      i) Real Time System
      ii) Time Sharing System.

   b) Explain any five file operations. **(6+6)**

3. a) Explain any six operating system services.

   b) Explain following Allocation Methods
      i) Contiguous allocation
      ii) Linked allocation. **(6+6)**

P.T.O.

Unit – II

4. a) What is paging ? Explain Paging Hardware with neat diagram.

b) Consider the following page reference string : 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 How many page faults would occur for the following replacement algorithm assuming three frames ?

i) LRU algorithm.

ii) Optimal replacement algorithm. **(6+6)**

5. a) Write a note on following :

i) Fragmentation

ii) Swapping.

b) What is segmentation ? Explain segmentation hardware with a neat diagram. **(6+6)**

Unit – III

6. a) What is process ? Explain process state transition with neat diagram.

b) List and explain various Scheduling Criteria. **(6+6)**

7. a) Consider the following set of processes, with length of the CPU-burst time given in milliseconds.

Process	**Burst time**
P1	15
P2	4
P3	10
P4	8
P5	5

Draw Gantt chart using round robin with time quantum of 5 milliseconds and find average waiting time.

b) Describe the differences between short-term, medium-term and long-term schedulers. **(6+6)**

Unit – IV

8. a) What is critical section ? What are the requirements for a solution to critical section problem ?

   b) What is deadlock ? What are the necessary conditions for deadlock situation to occur ? **(6+6)**

9. a) What is wait for graph ? Explain wait for graph with an example.

   b) Write a note on :

      i) Readers-Writers Problem

      ii) Dining Philosopher's Problem. **(6+6)**